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Coordinate or collaborate? Reducing food waste in perishable-product supply chains

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ABSTRACT

Reducing food waste in supply chains (SCs) with multiple decision-makers is challenging. A common approach grocery retailers use to reduce waste is requiring manufacturers to only send products with a long remaining shelf life ("minimum life on receipt"-MLOR). However, its impact on manufacturers remains unclear.

To evaluate the effectiveness of MLOR agreements on food waste, we investigate two strategies: (1) collaborating on setting the MLOR level and (2) coordinating the SC via contract. Through collaboration, we analytically show that if the MLOR agreement does not demand solely fresh products, it raises manufacturer profits, enabling potential wholesale price reduction. This might incentivize retailers to collaborate to reduce the MLOR level. We demonstrate that the coordinating strategy can reduce waste in the SC and is most beneficial when the wholesale price is high, and the issuing policy is FIFO. We introduce possible coordination contracts and show that in coordinated SCs, manufacturers always provide the highest MLOR level without requiring any restrictive MLOR agreements.

Governments mainly focus on reducing retail waste and promoting retailers to request higher MLOR. However, these efforts can backfire by creating more waste for manufacturers. Reducing the MLOR allows retailers to negotiate lower wholesale prices, increasing profitability while reducing waste. Although SC coordination is known for reducing inefficiency, it may not be the best strategy for reducing waste, especially when the issuing policy is more LIFO than FIFO. Specifically, while coordination might be a better strategy for online retailers, collaboration can be a better strategy for brick-and-mortar retailers.

1. Introduction

According to the United Nations Environment Programme's Food Waste Index Report 2021, grocery stores account for 13% of global food waste. This amounts to 121 million metric tons of food wasted in grocery stores, which is not trivial (UNEP, 2021). In particular, perishable food products are among the largest contributors to the global food waste problem (WRAP, 2015). As this high rate of food waste is increasingly seen as unacceptable due to its impact on the environment, social justice, and economy (FAO, 2017), food supply chains (SCs) are under pressure to manage and reduce their food waste more effectively. Retailers are equipped with considerable market power and can drive changes to curb food waste through their position in the supply chain, connected with upstream manufacturers and downstream customers. For example, a network of approximately 400 retailers, manufacturers, service providers, and stakeholders in 70 countries – with combined sales of 3.5 trillion euros – committed in 2015 to halve their food waste by 2030 (The Consumer Goods Forum, 2023).

However, managing the inventory of perishable products throughout the supply chain is more challenging than non-perishable products. Firstly, consumers seek products with long shelf lives. Empirical studies provide evidence that many consumers carefully screen products' expiry dates ("best before", "sell by", or "use by" dates) and pick the ones with the longest shelf life remaining (Tsiros & Heilman, 2005). Secondly, when ordering products from manufacturers, retailers also require long remaining shelf lives, in practice known as "minimum life on receipt", or MLOR (WRAP, 2015). A high MLOR level increases the time window for the retailer to sell the products, which reduces the retailers' waste. (Broekmeulen & van Donselaar, 2019) find that having just one more day of store shelf life results in 43.1% less waste on the retail side. However, a high level of MLOR puts more pressure on the manufacturer to deliver fresher products. Studies have found that, on average, retailers ask between 66% and 85% of the product's total shelf life as MLOR (European Commission, 2018; WRAP, 2015), implying an average internal shelf life of only 15% to 33% for handling,

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storage, and logistics. Such an "internal sell-by date" is one of the main causes of waste in the manufacturer stage (Garrone, Melacini, & Perego, 2014). An additional pressure for manufacturers is the high service level requirement of the retailer, which requires manufacturers to maintain substantial safety stocks and consequently account for a significant share of waste in the supply chain. While estimates seem to vary depending on the exact method and scope of study, the most recent academic work (Akkerman & Cruijssen, 2024) suggests that – apart from household food waste – most food waste seems to occur at the manufacturer, potentially driven by retailers having imposed increasingly tighter MLOR constraints.most Depending on the study, the contribution to food waste in the manufacturing and retail stage ranges between 14% manufacturing and 12% retail (Labs, 2022), 28% production and processing and 7% retail (Akkerman & Cruijssen, 2024), and 39% manufacturing and 5% wholesale/retail (EEA Signals, 2014).

One of the most cited causes of food waste is the lack of collaboration and coordination in supply chains (de Moraes, de Oliveira Costa, Roberta Pereira, da Silva, & Delai, 2020), mainly due to the existence of multiple opportunistically operating actors who base their decisions on different incentives (Govindan, 2018). The terms coordination and collaboration are used complementary to each other (Arshinder, Kanda, & Deshmukh, 2008). *Supply Chain Collaboration* is defined as a partnership process in which two or more autonomous companies work closely to plan and execute supply chain operations to achieve common goals and mutual benefits (Cao & Zhang, 2011). Following this definition, we examine a collaboration strategy, defined herein as the retailer granting a relaxation to the MLOR agreement and its impact on the retailer's and manufacturer's profits and waste. Despite MLOR agreements being widely used in practice, setting the level of MLOR is relatively understudied in academic research (Santos, Martins, Amorim, & Almada-Lobo, 2022). *Supply chain coordination* between interdependent organizations is a well-known approach that seeks to address the effect of double marginalization using different mechanisms, such as contracts, information exchange, and joint decision-making (Cachon, 2003). Efficiency and performance are mainly related to profitability or cost criteria, and a supply chain is called coordinated when decisions in a decentralized supply chain (SC) lead to the same performance as in a centralized one (Arshinder et al., 2008; Lariviere, 1999). Although various incentive mechanisms have been studied to coordinate supply chains of non-perishable products, researchers have not yet evaluated the impact of perishability and waste. Most supply chain coordination studies assume an infinite capacity at the supply stage and a single opportunity to order inventory (Cachon, 2003 defines these as single-location problems). However, the perishable nature of a product and the possibility of replenishing inventory require a multi-period, two-location problem, being more challenging since the infinite capacity at the upstream manufacturing stage cannot be assumed (Cachon, 2003).

Following (Akkaş & Gaur, 2022) who call for more research that incorporates food waste into retailing decisions and specifically highlight the importance of providing incentives for independent firms to cooperate in reducing waste, in this study we seek to address the inefficiency in perishable-product supply chains by focusing on two possible strategies: (1) collaboration between retailers and manufacturers on setting the required MLOR level and (2) coordination of the decision-making processes of retailers and manufacturers through a coordinating contract. We consider a two-echelon decentralized SC consisting of one retailer and one manufacturer selling a perishable product with a finite shelf life, both maximizing their expected profit and measuring their waste. It encompasses food products for which the manufacturer and the retailer guarantee constant quality within the finite shelf life, for example, for prepackaged food such as dairy products (milk, cheese, yogurt) or pre-cooked meals. For such products, retailers typically keep products with several remaining "best before" or "sell by" dates, which, in turn, depend on the manufacturer's production time.

Regarding the collaboration strategy, we investigate a case in which the retailer and the manufacturer collaborate on setting the MLOR level. Our analysis shows that when the MLOR level is reduced, the manufacturer's waste and costs decrease, allowing him to accept a lower wholesale price while making the same profit and increasing the retailer's profit. Therefore, the collaboration strategy reduces the total SC's waste alongside a surprising profit increase.

To examine SC coordination, we define a centralized decision-maker model as a benchmark to compare it with the decentralized SC setting. We numerically illustrate that coordinating the SC creates a clear advantage in reducing waste and increasing profit in the SC. The issuing policy and the wholesale price in the baseline scenario heavily influence the extent of this impact. Furthermore, we emphasize that a coordinated SC ensures that the highest possible MLOR level is met since the manufacturer always supplies freshly produced products.

Comparing waste and profit numerically, our results show that while a coordinated SC is always better off concerning profit, this is not the case concerning waste. The sensitivity of waste related to changes in the issuing policy (first-in-first-out (FIFO), last-in-first-out (LIFO), or a hybrid form) is significantly higher. Thus, when retailers have less control over issuing policy (like brick-and-mortar retailers), the collaborated SC can achieve lower waste than the coordinated one. This is interesting since, in the last decade, the literature has developed many mechanisms, such as vendor-managed inventory, to increase SC efficiency. The coordinated SC might be a better choice when retailers have more control over the issuing policy (like online grocery retailers) since it provides a higher profit under similar waste results. It is crucial to note that the side benefit of coordinating the SC for retailers is enabling them to receive the highest MLOR level (freshness level) without imposing any MLOR agreement on their manufacturers.

Furthermore, we design coordination contracts for perishable-product supply chains and show that coordination is possible when the retailer and manufacturer have scanned-based trading or private-label production agreements. In the former, coordination is possible through a two-part tariff contract with a zero wholesale price and a fixed side payment. In contrast, in the latter, coordination is achievable via various commonly known contracts, such as revenue-sharing contracts.

The remainder of this article is organized as follows: In Section 2, we review the related literature and position our paper. In Section 3, we develop the mathematical model for the respective objectives of the retailer and the manufacturer. In Section 4, we discuss the collaboration strategy and the effect of MLOR agreements on the waste and profit of SC members. In Section 5, we study the coordination strategy and its impact on SC waste and profit. In Section 6, we discuss possible contracts coordinating SC of perishable products. In Section 7, we numerically compare coordination and collaboration strategies under different conditions. We conclude this study in Section 8.

2. Literature review

Our research relates to two streams of literature, *inventory control of perishable products* and *food waste in sustainable operations management*.

There is extensive literature that studies the management of perishable inventory. Therefore, we narrow it down to the most related stream of literature. Goyal and Giri (2001) identified two classes of perishable inventory: (i) inventory that becomes obsolete at the end of the lifetime due, for example, to new products, technologies, or getting out-of-fashion, and (ii) inventory that physically deteriorates due to spoilage, damage, dryness, etc. and has to be reordered to meet future demand. Deteriorating inventory can be further classified according to their shelf life characteristics: (a) products with fixed (predetermined) shelf life, (b) products with an age-dependent deterioration rate, and (c) products with deterioration dependent on time or inventory (Bakker, Riezebos, & Teunter, 2012). Our paper belongs to class (ii, a), i.e., we consider a deteriorating product with a fixed shelf life. Comprehensive literature reviews are provided by Bakker et al. (2012), Goyal and Giri (2001), Janssen, Claus, and Sauer (2016), Karaesmen, Scheller-Wolf, and Deniz (2011) and Nahmias (2011). We furthermore examine a

perishable inventory control problem in a multi-echelon context. Thus, we restrict our review to research that provides insight into multi-echelon perishable product SCs, which have proven to be particularly complex.

Many scholars have examined multi-echelon problems from the perspective of a central planner. In an early study, Federgruen, Prastacos, and Zipkin (1986) investigate an allocation and distribution problem of a system with a single warehouse and multiple retailers, in which the allocation of age-dependent stock is considered. Goh, Greenberg, and Matsuo (1993) study a two-stage system in which fresher products are stocked at the upstream stage and older products at the downstream stage (typically relevant for blood banks). Highlighting the complexity of such a problem, the authors develop two heuristics. Fujiwara, Soewandi, and Sedarage (1997) consider an ordering and issuing problem in a two-stage system where products have different shelf lives depending on whether they are stocked in the first stage (e.g., cool-room) or in the second stage (e.g., supermarket shelves). When studying decision-making processes in multi-echelon SCs of perishables, scholars typically simplify their work by using deterministic models, e.g., Lee, Rhee, and Cheng (2013) and Zhang, Luo, and Duan (2016). Other studies have used approximations that allow them to apply a single-period formulation. For example, Ketzenberg and Ferguson (2008) consider an uncertain remaining lifetime upon receipt in the downstream stage of a two-echelon SC to investigate the value of information sharing. They show how information sharing increases product freshness and decreases outdating within the SC. Meanwhile, Minner and Transchel (2017) examine the impact of perishability on the bullwhip effect in SCs by analyzing the variability of a retailer's order.

While it is common for retailers to require a certain level of remaining shelf life (MLOR), only a few articles have studied the (remaining) shelf life issues such as MLOR as a decision variable. Pauls-Worm, Hendrix, Alcoba, and Haijema (2016) study the impact of internal shelf-life settings in a centralized two-stage production–inventory control problem and show that a higher internal shelf life at the production stage reduces waste levels at the producer. Santos et al. (2022) study a similar problem. For an integrated production–inventory control problem in a two-stage perishable-product supply chain, (Santos et al., 2022) show that relaxing the MLOR agreement by 12% only reduces the average freshness of products at a retailer by 1%. Akkaş and Honhon (2022a) optimize the MLOR level for a centralized two-stage SC of an industry collaborator and illustrate that an optimal MLOR level provides the potential to reduce waste and increase profits by almost 15% and 9%, respectively.

Our study differs from previous MLOR studies by investigating a two-stage SC where the downstream stage (retailer) and the upstream stage (manufacturer) are two independent decision-making units that maximize their own profit. We provide analytical insights where possible. It is crucial to note that an optimal decision of one firm may not be in the best interest of the overall SC, which can lead to severe SC inefficiencies (Cachon, 2003). Therefore, we investigate possible coordination contracts for such a setting in more detail. In addition, we investigate different inventory issuing policies (first-in-first-out (FIFO), last-in-first-out (LIFO), or a hybrid form), which have been shown to be critical in perishable product inventory control (Cohen & Prastacos, 1981; Nahmias, 2011).

We also contribute to the literature on *sustainable operations management focusing on the reduction of food waste*. Most studies in this field look at either the SC's downstream or upstream (Ata, Lee, & Sönmez, 2019) focus on food waste on farms to improve operations to increase the volume of fresh food produced. Akkaş and Sahoo (2020) show that manufacturers can reduce waste by aligning their incentives to the sales force. Akkaş and Honhon (2022b) analyze the impact of warehouses' shipment policies (including the age composition of inventories) on profit and waste. Having said that, some other studies have looked into the food waste of SCs downstream. For instance, Belavina, Girotra, and Kabra (2017) study the waste on the consumer side, showing that subscriptions reduce food waste due to customers making more frequent and smaller orders. Akkaş (2019) highlights the importance of enforcing inventory rotation on retail stores' shelves and its impact on food waste. Looking at product shelf lives, it has been shown that increasing the freshness or sales reduces the waste in grocery retailers (Broekmeulen & van Donselaar, 2019). Akkaş, Gaur, and Simchi-Levi (2019) investigate the main drivers of product expiration in the consumer packaged goods industry. Belavina (2021) study the effect of the density of grocery stores on the waste generated by the stores themselves and their final consumers. Finally, (Akkaş & Gaur, 2022) highlight the importance of studying the reduction of food waste at all levels of the SC. One of their suggested approaches for dealing with food waste is incentivizing SC members to cooperate.

We contribute to the literature on food waste by examining the possibility of reducing food waste when SC members either collaborate on setting the MLOR level or use contracts to coordinate the entire SC. Some studies address coordination mechanisms for multi-period problems where the retailer has multiple replenishment opportunities. Our paper focuses on a two-echelon problem setting for perishable products, where the manufacturer might not always be able to fulfill the orders in full. We aim to identify the impact of coordination contracts and collaboration over the MLOR level on the profit and waste of a two-location SC (consisting of one retailer and one manufacturer). The manufacturer faces a positive production lead time, and for both parties, unsatisfied demand will be lost. To our knowledge, a very small number of studies have tracked the age of products over the contract horizon or considered the possibility of receiving products from manufacturers with different age categories.

3. Model description

We consider a two-echelon SC with one manufacturer (indexed by m and denoted as "he") and one retailer (indexed by r and denoted as "she"). The supply chain sells a single product with a finite shelf life of N periods, i.e., together with the time of production, N directly determines the known "best before" or "sell by" date, which consumers see when they buy the product. We consider a food product for which the manufacturer (and, respectively, the retailer) guarantee a constant quality within the N period shelf life, as it is, for example, for prepackaged food such as dairy products (milk, cheese, yogurt) or precooked meals. The selling price, r, does not depend on the remaining shelf life (similar to, Chung & Erhun, 2013; Li, Lim, & Rodrigues, 2009). At the end of the period N, the product will be removed from the shelf, and sent for donation to organizations such as food banks or considered waste. Note that we assume that the retailer does not apply any markdown pricing of end-of-shelf-life products, a practice of various retailers in Europe and other parts of the world.

3.1. Stage 1: Contracting

In line with grocery retail, we presume a power imbalance between the retailer and manufacturer. To do so, we use a Stackelberg game to describe their relationship, where the retailer (as the Stackelberg leader) offers a contract, Ω, to the manufacturer as a take-it-or-leave-it proposition. The contract specifies the wholesale price w and the MLOR level $n \leq N$, i.e., $\Omega = \Omega(w, n)$. The manufacturer (as a Stackelberg follower) accepts the agreement when his long-run expected profit per period, $\hat{\Pi}_m(\Omega, \mathbf{p}^*(\Omega))$, following the optimal production policy $\mathbf{p}^*(\Omega)$, is above his reservation profit, π_m. It is assumed that π_m is known by the retailer. We require that the outcome of the game to be a subgame perfect equilibrium, where the retailer optimizes her strategy (decision) given that the manufacturer adheres to a specific strategy (decision) (Romp, 1997). Thus, the retailer maximizes her long-run expected profit per period by optimizing the contract (Ω) while anticipating to follow the optimal ordering policy $\mathbf{q}^*(\Omega)$ and ensuring that the manufacturer receives at least his reservation profit, i.e.,

$$\max \hat{\Pi}_r(\Omega, \mathbf{q}^*(\Omega)) \quad s.t. \quad \hat{\Pi}_m(\Omega, \mathbf{p}^*(\Omega)) \geq \pi_m. \tag{1}$$

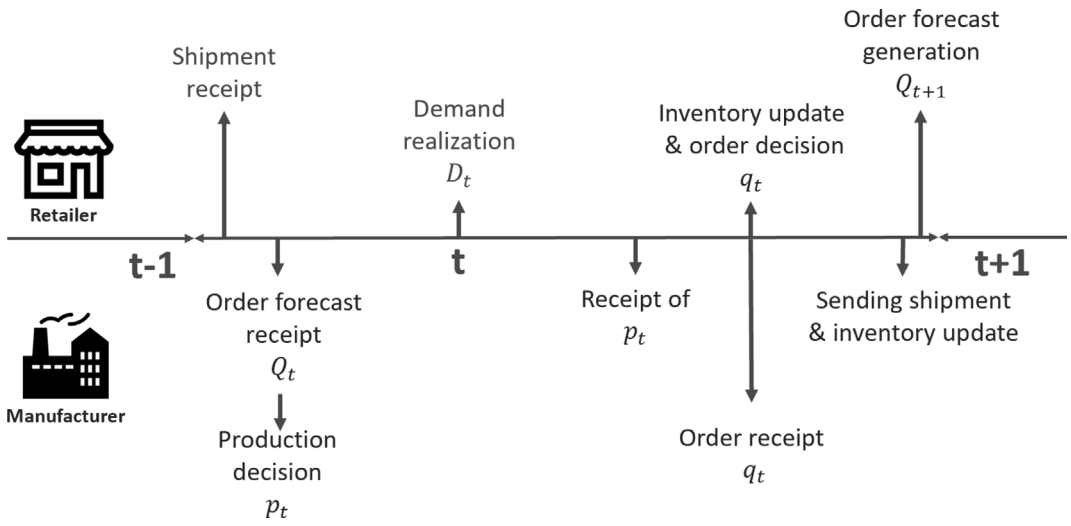


Fig. 1. The sequence of events for the retailer and manufacturer in each period.

3.2. Stage 2: Operational stage

At the operational stage, the retailer and manufacturer periodically review their inventories and make their orders and production decisions, respectively, following a myopic policy. That is, in every period t, the manufacturer makes his production decision, p_t at the beginning of period t given the order forecast (or similar advanced order information), Q_t received from the retailer in the previous period. Q_t is technically the demand of the manufacturer's demand. During period t, the retailer observes demand D_t, updates her inventory levels, and decides on the order quantity q_t. Finally, the retailer determines a forecast of the order placed in period $t+1$, Q_{t+1}. Both q_t and Q_{t+1} are sent to the retailer. The manufacturer then seeks to fulfill the retail order q_t as good as possible and the profits of both retailer and manufacturer are realized. The objective of retailer and manufacturer is to maximize the average long-run expected profits for a given contract Ω throughout the contract horizon T, which are formalized by:

$$\hat{\Pi}_r(\Omega, \mathbf{q}^*(\Omega)) = \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \Pi_{r,t}(q_t^*|\Omega), \quad \text{and}$$
$$\hat{\Pi}_m(\Omega, \mathbf{p}^*(\Omega)) = \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \Pi_{m,t}(p_t^*|\Omega). \tag{2}$$

$\Pi_{r,t}(q_t^*|\Omega)$ and $\Pi_{m,t}(p_t^*|\Omega)$ are the expected profits of the retailer and manufacturer in period t given their optimal ordering and production policy, respectively, for a given contract Ω. In the following, we discuss the objective function and the sequence of events for both the retailer and the manufacturer in more detail, as illustrated in Fig. 1.

3.2.1. Retailer

The retailer follows a dynamic replenishment strategy. At the beginning of every period t, she receives the shipment of her previous order q_{t-1}, which could not be fulfilled in full when the manufacturer does not have enough inventory. However, we show later that by a sufficiently high service level agreement between the retailer and the manufacturer, this problem can be eliminated. After observing the actual demand, the retailer updates her inventory and removes products with zero shelf life. At the end of every period, she makes an ordering decision, (q_t), such that it maximizes her expected profit of period t_{t+1}. The next period demand D_{t+1} is random with the probability density function (p.d.f.) $f_{t+1}(\cdot)$ and cumulative distribution function (c.d.f.) $F_{t+1}(\cdot)$. The retailer's objective function in every period t is:

$$\Pi_{r,t}(q_t) = r\mathbb{E}\left(\min\left\{D_{t+1}, q_t + \sum_{j=1}^{N-1} y_{r,t}^j\right\}\right) - \beta_r \mathbb{E}\left(D_{t+1} - q_t - \sum_{j=1}^{N-1} y_{r,t}^j\right)^+ - wq_t \tag{3}$$

where $y_{r,t}^j$ is the inventory of the retailer with a remaining shelf life of j periods at the end of period t. The first and last terms in Eq. (3) are the expected revenue and the purchase cost with a wholesale price of w, respectively. The second term is the expected cost of goodwill with β_r as a penalty for each unit of demand that the retailer does not satisfy. Since the cost of holding inventory is rather low, we ignore the holding cost, a common assumption in perishable-product inventory control (e.g., see Chung & Erhun, 2013; Haijema & Minner, 2016; Zhou, Leung, & Pierskalla, 2011). This simplification allows us to avoid unnecessary complications that do not serve the main purpose of this study. The objective function introduced in (3) has the classical newsvendor form, and by getting a derivative from it with respect to q_t, the optimal ordering decision in period t is:

$$q_t^* = \left(F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) - \sum_{j=1}^{N-1} y_{r,t}^j\right)^+. \tag{4}$$

The inventory levels $y_{r,t}^j$ in each age category j following the respective inventory issuing policy (FIFO or LIFO). Under FIFO, the inventory transition equations are as follows:

$$y_{r,t}^j = \left[y_{r,t-1}^{j+1} + \eta_{t-1}^{j+1} - \left(d_t - \sum_{i=1}^{j-1}(y_{r,t-1}^{i+1} + \eta_{t-1}^{i+1})\right)^+\right]^+, \forall j = 1, \ldots, N-2.$$

and

$$y_{r,t}^{N-1} = \left[\eta_{t-1}^{N} - \left(d_t - \sum_{i=1}^{j-1}(y_{r,t-1}^{i+1} + \eta_{t-1}^{i+1})\right)^+\right]^+,$$

where η_{t-1}^j represents the shipment sent with the remaining shelf life of j at the end of period $t-1$. Under LIFO, the transition equations change only slightly.

$$y_{r,t}^j = \left[y_{r,t-1}^{j+1} + \eta_{t-1}^{j+1} - \left(d_t - \sum_{i=j+1}^{N-1}(y_{r,t-1}^{i+1} + \eta_{t-1}^{i+1})\right)^+\right]^+, \forall j = 1, \ldots, N-2$$

$$y_{r,t}^{N-1} = \left(\eta_{t-1}^{N} - d_t\right)^+.$$

Dependency to manufacturer's production decision: The retailer places the order q_t^* with the manufacturer at the end of period t that the manufacturer has to serve immediately. The retailer determines at the same time an order forecast, Q_{t+1}, i.e., a prediction of the order placed at the end of the next period $t+1$. Since we consider a production

lead time of one period, this order forecast Q_{t+1} is a random variable with a known distribution that serves as a demand forecast for the manufacturer in period $t+1$ and is formalized as follows:

$$Q_{t+1} = \left(F^{-1}\left(\frac{r+\beta_r - w}{r+\beta_r} \right) - D_{t+1} - Y^0_{r,t+1} \right)^+. \tag{5}$$

$Y^0_{r,t+1}$ denotes the random inventory level at the end of period $t+1$with zero periods remaining shelf life. Depending on the retailer's inventory issuing policy (FIFO or LIFO), $Y^0_{r,t+1}$ is calculated by

$$Y^0_{r,t+1} = \begin{cases} \left(y^1_{r,t} - D_{t+1} \right)^+ & FIFO \\ \left(y^1_{r,t} - \left(D_{t+1} - \sum_{j=2}^{N-1} y^j_{r,t} \right)^+ \right)^+ & LIFO \end{cases}. \tag{6}$$

From Eqs. (5) and (22), we can see that the retailer's order quantity for the next period $t+1$, derived at the end of period t, is a random variable that depends not only on the demand D_{t+1} and the total inventory at the end of period t but also on the amount of inventory with only a one-period remaining shelf life. The corresponding p.d.f. and c.d.f. of Q_{t+1} is denoted by $g_{t+1}(\cdot)$ and $G_{t+1}(\cdot)$, respectively. The derivation of Q_{t+1}, including its density and distribution functions, is illustrated in Appendix A.

3.2.2. Manufacturer

The manufacturer has a production lead time of one period. Note that under zero production lead time, the decision problem would be trivial because the manufacturer could produce the retailer's order exactly without any waste or missed inventory. The sequence of events for the manufacturer in period t is illustrated in Fig. 1 and is as follows: (1) At the beginning of a period, the manufacturer receives the retailer's order forecast, Q_t (as a random variable with a distribution characterized by p.d.f. $g_t(\cdot)$ and c.d.f. $G_t(\cdot)$) and makes his production decision p_t; (2) given the one-period production lead time, p_t will be available at the end of the same period; (3) after receiving the retailer's order, q_t, at the end of a period the manufacturer fulfills it with the available inventory based on the required MLOR level, n, to the greatest extent possible, and finally (4) the inventory level is updated by removing products that do not meet the MLOR requirement in the next period. We assume that unmet demand from the retailer is lost. This is typical in grocery retail where products are frequently replenished, see, e.g., UNFI (2024).

The manufacturer makes his production decision at the beginning of a period to maximize his expected profit by fulfilling the retailer's order at the end of the period, subject to a service level requirement α, that characterizes the non-stockout probability. The objective function is as follows:

$$\Pi_{m,t}(p_t) = w\mathbb{E}\left(\min\left\{ Q_t, p_t + \sum_{j=n+1}^{N} y^j_{m,t-1} \right\} \right) - cp_t \tag{7}$$

$$s.t.\ Pr\left(Q_t \le p_t + \sum_{j=n+1}^{N} y^j_{m,t-1} \right) \ge \alpha \tag{8}$$

Eq. (7) refers to the manufacturer's expected revenue, depending on the wholesale price w, which is part of the contract Ω minus the production cost with a unit cost of c ($0 < c < w < r$). Eq. (8) refers to the service level requirement of the retailer. The optimal production decision also depends on the current inventory state at the end of period $t-1$, defined by $y^j_{m,t-1}$, where j is the remaining shelf life of the products. Due to the MLOR agreement, the manufacturer cannot send products with a remaining shelf life of less than n periods. Thus, the optimal time-dependent production decision (p^*_t) is:

$$p^*_t = \max\left\{ \left(G^{-1}_t(\alpha) - \sum_{j=n+1}^{N} y^j_{m,t-1} \right)^+, \left(G^{-1}_t\left(\frac{w-c}{w} \right) - \sum_{j=n+1}^{N} y^j_{m,t-1} \right)^+ \right\} \tag{9}$$

At the end of period t, the manufacturer receives the retailer's order quantity q^*_t that he needs to fulfill. In a B2B context, it is valid to assume that the manufacturer's issuing policy is a pure FIFO policy. Hence, the manufacturer fulfills the order from his inventory, given by the inventory vector from period $t-1$, $(y^{n+1}_{m,t-1}, y^{n+2}_{m,t-1}, \dots, y^N_{m,t-1} = p^*_{t-1})$ as shown as follows:

$$\eta^n_t = min\left\{ y^{n+1}_{m,t-1}, q^*_t \right\}$$

$$\eta^j_t = min\left\{ y^{j+1}_{m,t-1}, \left(q^*_t - \sum_{i=n}^{j-1} y^{i+1}_{m,t-1} \right)^+ \right\}, \forall j = n+1, \dots, N-1$$

$$\eta^N_t = min\left\{ p^*_{t-1}, \left(q^*_t - \sum_{i=n}^{N-1} y^{j+1}_{m,t-1} \right)^+ \right\}.$$

Theorem 1. *When $\alpha \ge \frac{r+\beta_r-w}{r+w}$, the manufacturer can always fulfill the complete order of the retailer.*

At the end of each period, the manufacturer updates his inventory level for each age category as follows:

$$y^j_{m,t} = y^{j+1}_{m,t-1} - \eta^j_t, \quad \forall j = n+1, \dots, N-1$$

$$y^N_{m,t} = p^*_t - \eta^N_t.$$

In the next two sections, we investigate how collaboration over setting the MLOR level or coordinating the SC can impact the SC's waste and profit.

4. Collaborating in the SC

In this section, we analyze the effect when the retailer and the manufacturer collaborate to set the MLOR level. To provide analytical results, we first establish a baseline scenario in which the retailer requires the manufacturer to send only products with the highest MLOR level, i.e. ($n = N$) and compare it with a more collaborative case where the retailer accepts products that are at most one period old ($n = N-1$).

Waste perspective

To evaluate the effect of MLOR collaboration on waste in the SC, we need to determine how waste will be influenced on the retail side ($\theta_{r,t}$) and manufacturing ($\theta_{m,t}$) side in every period by manipulating the MLOR level. One of the elements that affects waste is the issuing policy. For the manufacturer, we adopt a FIFO issuing policy since it is common in practice that manufacturers have control over their issuing policy and it is shown to be the optimal issuing policy (Akkaş & Honhon, 2022b). Unlike the manufacturer, the retailer's issuing policy depends on the customer's purchasing behavior and the rotation of older inventory on the store's shelves. Although, in practice, the retailer's issuing policy is often a mixture of LIFO and FIFO, in this section, we consider the LIFO issuing policy for the retailer. By considering FIFO for the manufacturer and LIFO for the retailer, we aim to highlight the importance of our results under these relatively pessimistic assumptions. Later, we will conduct a sensitivity analysis of the retailer's issuing policy using our numerical results.

Manufacturer: At the end of each period t, the manufacturer removes all the products with n periods remaining shelf life. Thus, the expected waste of the manufacturer at the end of period t is:

$$\theta_{m,t} = \mathbb{E}\left(y^{n+1}_{m,t-1} - Q_t \right)^+. \tag{10}$$

For a specific case, when the manufacturer is only required to send the freshest products ($n = N$), the expected waste of the manufacturer is:

$$\theta_{m,t} = \mathbb{E}(p^*_t - Q_t)^+. \tag{11}$$

The long-run expected waste per period is $\theta_m(w,n) = \lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \theta_{m,t}(w,n)$. The following theorem illustrates the change in the manufacturer's long-run expected waste under the two MLOR agreements, i.e., $n = N$ and $n = N-1$.

Theorem 2. *By reducing the MLOR level from $n = N$ to $n = N-1$, it follows that*

(a) $\theta_m(w, N-1) < \theta_m(w, N)$,
(b) The reduction in the expected waste per period of the manufacturer is $\frac{1}{2}\left((p_t^*)^2 - (y_{m,t-1}^N)^2\right)$.

Theorem 2 shows that by reducing the MLOR level by one period, and thus providing the manufacturer with more flexibility in managing his inventory, the manufacturer's expected waste per period reduces by $\frac{1}{2}((p_t^*)^2 - (y_{m,t-1}^N)^2)$ given the FIFO issuing policy.

Retailer: The retailer's expected waste in each period under the LIFO issuing policy is:

$$\theta_{r,t} = \mathbb{E}\left(y_{r,t}^1 - \left(D_{t+1} - q_t^* - \sum_{j=2}^{N-1} y_{r,t}^j\right)^+\right)^+. \tag{12}$$

Note that the MLOR agreement directly affects the age categories of the arriving orders ($q_t^*(w,n)$). When $n = N$, the retailer only receives products with the longest remaining shelf life of N periods ($q_t^*(w,N) = y_{r,t}^N$). However, when $n = N-1$, the retailer might also receive products with $N-1$ periods of remaining shelf life ($q_t^*(w,N-1) = y_{r,t}^N + y_{r,t}^{N-1}$).

Theorem 3. *By reducing the MLOR level from N to $N-1$, the increase in the retailer's average expected waste per period is strictly less than* $\frac{q_t^*}{N(N-1)}$.

Theorem 3 shows that by reducing the MLOR by one level, although the average expected waste per period of the retailer increases, it is bounded. Comparing the results of Theorems 3 and 2(b), if the N is sufficiently large, then by switching to a lower MLOR level, the total waste in the SC decreases.

We conducted a detailed numerical study to examine the sensitivity of our results to various parameters: cost of goodwill, maximum shelf life, demand uncertainty, the manufacturer's service level requirement, and the retailer's issuing policy. We assume a product with a unit retail price of $r = 11$ and a unit production cost of $c = 7$ to ensure positive profits for both the retailer and the manufacturer. The numerical results (except Fig. 4) are the average over possible wholesale prices ($w = 7, 7.5, \ldots, 11$) to ensure the results are independent of w. The maximum shelf life varies between $N = 5$ and $N = 15$ periods. The demand is assumed to follow a negative binomial distribution with a mean $\mu = 10$. We compare a scenario with low demand uncertainty where the standard deviation is $\sigma = 5$ and a scenario with high demand uncertainty in which $\sigma = 10$. Smith and Agrawal (2000) find that the negative binomial demand distribution performs well in fitting retail sales data. We further study the effect of customers' goodwill cost by assuming $\beta_r = 5$ and $\beta_r = 15$ and the effect of service level requirements for the manufacturer by assuming no additional service level requirement and $\alpha = 0.95$. The initial inventory of the retailer (y_r^N) and manufacturer (y_m^N) is equal to 5. To find an average long-run expected profit, we set the planning horizon T at 1000 periods.

To quantify the waste, we define a waste index that normalizes the long-run average expected waste of the retailer (θ_r) and manufacturer (θ_m) to the mean of final consumer demand:

$$WasteIndex_m = \frac{\theta_m(w,n)}{\mu}, \qquad WasteIndex_r = \frac{\theta_r(w,n)}{\mu} \tag{13}$$

Figs. 2 and 3 illustrate the effect of an MLOR relaxation on the manufacturer's and the retailer's waste index for different levels of (a) cost of goodwill, (b) maximum shelf life, (c) demand uncertainty, (d- Fig. 2) service level requirement of the manufacturer, and (d- Fig. 3) the retailer's issuing policy.

Manufacturer: Fig. 2 shows that for all scenarios, a lower MLOR level significantly lowers the manufacturer's waste index. Concerning sensitivity to individual parameters, we find that when the retailer has a higher cost of goodwill (β_r), the manufacturer ends up with more waste under $n = N$ (Fig. 2(a)). The reason is that, under higher goodwill costs, the retailer increases the order size to avoid unwanted stockout costs, which consequently leads to an increase in the manufacturer's production quantity and waste. When the retailer requires the manufacturer to send only the freshly produced products, all of the manufacturer's excess inventory will be wasted at the end of the period. Thus, having a higher cost of goodwill on the retail side increases the manufacturer's waste. Therefore, for products with higher goodwill costs, it is more important to require a lower level of MLOR to reduce waste at the manufacturer.

Fig. 2(b) shows that the shelf life of the product plays a minor role with respect to the benefits of an MLOR reduction. The change of the manufacturer's waste is rather identical for $N = 5$ and $N = 15$. Fig. 2(c) illustrates that although having higher demand uncertainty (standard deviation) results in higher wastes under any specific MLOR level, reducing the MLOR lever always results in less waste at the manufacturer's facility. Fig. 2(d) compares the manufacturer waste under no specific service level (α) requirement and a high service level requirement. This figure highlights that switching to a lower MLOR level is more beneficial under a high required service level, which is a common practice in the grocery industry.

Retailer: Fig. 3 reveals that for all scenarios, retailer waste increases negligibly when the MLOR level is reduced from $n = N$ to $n = N-1$. Although one might expect that reducing the MLOR level even further (to $n = N-2$ or $n = N-3$) would increase the retailer's waste even more since the retailer agrees to receive fewer fresh products, this is not necessarily the case. Requiring a lower MLOR level only gives the manufacturer more flexibility to fulfill the orders. However, the manufacturer might not even have that many "old" products in stock to send to the retailer because he optimizes the production quantity in every period based on his available inventory.

Fig. 3(a) and (c) show that goodwill costs and increasing demand uncertainty (standard deviations) change the MLOR effect only negligible, which is a similar effect as for the manufacturer. Comparison of Figs. 2(b) and 3(b) for products with a long shelf life, that is, $N = 15$, shows that while switching to a lower MLOR level does not impact the retailer's waste, it has a significant impact on the manufacturer's waste. Fig. 3(d) shows the impact of different issuing policies (FIFO, mixed policy, and LIFO) on the retailer's waste. We previously assumed that the retailer's issuing policy is LIFO, knowing that this is a pessimistic assumption. Fig. 3(d) shows that even under a pure LIFO policy, reducing the MLOR level leads to a minor increase in retailer waste. Therefore, the waste in the SC decreases when the manufacturer's required MLOR level is lower.

Profit perspective

The manufacturer makes his production decision based on the wholesale price (w) and the MLOR level (n), and maximizes his expected long-term profit $\hat{\Pi}_m^*(\mathbf{p}^*(w,n)|(w,n)) = \lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\Pi_{m,t}(p_t^*|(w,n))$. Theorem 4 shows how a lower MLOR level impacts the manufacturer's production decision and the long-run average expected profit.

Theorem 4. *By reducing the MLOR level from $n = N$ to $n = N-1$, it follows:*

(a) For each given period t, $p_t^*(w, N-1) \le p_t^*(w, N)$,
(b) $\hat{\Pi}_m(\mathbf{p}^*(w, N-1)|(w, N-1)) > \hat{\Pi}_m(\mathbf{p}^*(w, N)|(w, N))$, and
(c) For the optimal wholesale price at the first stage it holds $w^*(N-1) < w^*(N)$.

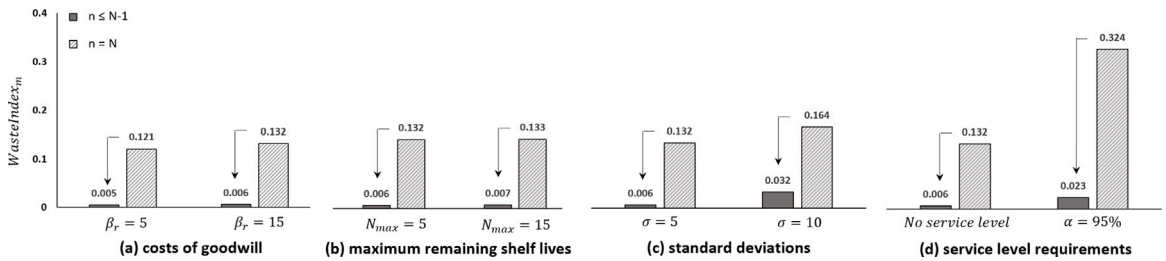


Fig. 2. Changes in the manufacturer's waste by reducing the MLOR level under different.

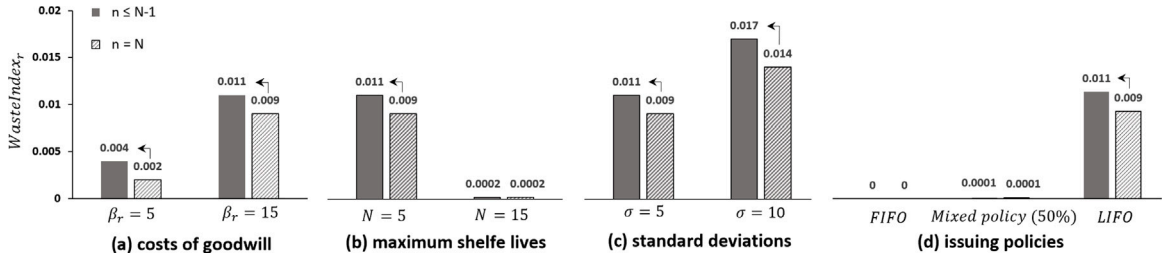


Fig. 3. Changes in the retailer's waste by reducing the MLOR level under different.

Theorem 4(a) indicates that a higher MLOR level leads to an increase in production quantity in every period t. The intuition is that a higher MLOR level leads to a shorter effective shelf life for the manufacturer. Therefore, the manufacturer needs to produce more to maximize his expected profit. Theorem 4(b) shows that a lower MLOR level impacts the manufacturer's long-run average profit positively. Since a lower MLOR level implies a longer effective shelf life for the manufacturer, more inventory can be carried over to the next period. This implies a lower production quantity, lower production costs, and thus, a higher long-run average profit (given the same wholesale price). Theorem 4(c) illustrates the MLOR effect on the optimal wholesale price, $w^*(n)$. Given an MLOR level $n = N$ for which the retailer, acting as a Stackelberg leader, optimizes the wholesale price $w^*(N)$ so that the manufacturer exactly earns his reservation profit (π_m). Together with the finding of Theorem 4(b), we find that by lowering the MLOR level to $n = N - 1$, the retailer can find a lower wholesale price, $w^*(N-1) < w^*(N)$, which guarantees the same reservation profit π_m for the manufacturer. This reduction in the wholesale price increases the retailer's profit without changing the manufacturer's profit.

Fig. 4 continues the numerical example that illustrates changes in profits for both the manufacturer and the retailer at various wholesale prices and different MLOR agreements. In the lower graphs of Fig. 4, showing the expected profits of the manufacturer for both $n = N$ and $n = N - 1$, it is evident that for any given reservation profit of the manufacturer (represented by any horizontal line), there exist two potential wholesale prices. A higher wholesale price corresponds to a higher MLOR level, while a lower wholesale price corresponds to a lower MLOR level. The upper graph in Fig. 4 shows the expected profit of the retailer for both $n = N$ and $n = N - 1$ (almost identical profits) and demonstrates that by transitioning to a lower MLOR level, the retailer can negotiate a reduced wholesale price while ensuring that the manufacturer still achieves his reservation profit. This adjustment allows the retailer to benefit from a higher profit margin resulting from the lower wholesale price, leading to a higher profit. A comparison between two reservation profits ($\pi_m = 25$ and $\pi_m = 35$) highlights that in scenarios where the manufacturer requires a higher reservation profit, collaborative efforts to decrease the MLOR level prove to be more advantageous for the retailer. It should be noted that for any specific manufacturer reservation profit, a potential lower wholesale price exists if the retailer is open to collaborating on reducing the MLOR level.

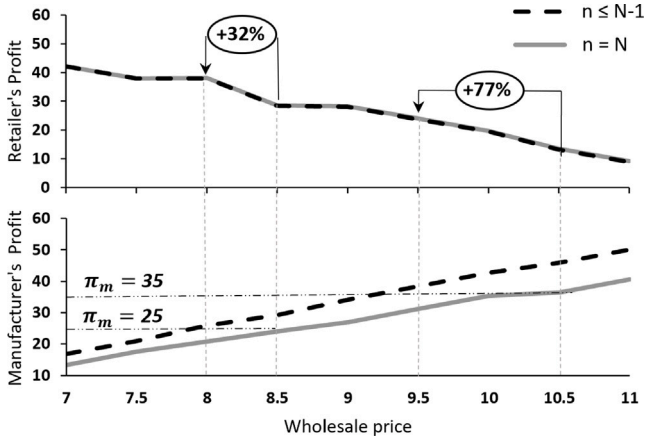


Fig. 4. The wholesale price change and the retailer's profit increase by requiring a lower MLOR level under different reservation profits π_m.

5. Coordinating the SC

Supply chain coordination is a well-known approach to improving an SC's efficiency (Cachon, 2003). A supply chain is considered coordinated when decisions in a decentralized SC align with those of a centralized one and achieve an equivalent level of performance (Lariviere, 1999). Although many studies in the coordination literature have focused on improving the SC's total profit, none have considered the effect of coordination on the MLOR agreement and the total supply chain waste. We noted in Section 3 that due to the positive lead time and lost sale settings, it is highly unlikely to find the overall optimal decisions for the SC with the traditional steady-state analysis. As a solution to this issue, we propose a more restrictive coordination definition for such a multiperiod setting.

Definition 1. *In a multiperiod problem setting, a decentralized SC is considered coordinated when the SC is coordinated during each period under study.*

Based on Definition 1, if a contract design provides the decentralized SC with identical decisions as our centralized decision-maker in any period (t), we can claim that this contract coordinates the

SC throughout the contract horizon (T). We are interested in finding the optimal decision of the centralized decision-maker in each period as the coordinated SC benchmark and comparing it with our previously described baseline scenario in which the retailer requires the manufacturer to send only the freshly produced products ($n = N$).

Centralized decision-maker

To find the objective function of the centralized decision-maker, it is common to sum the objectives of the retailer and the manufacturer (see e.g., Cachon, 2003; Ho & Zhang, 2008; Sieke, Seifert, & Thonemann, 2012). However, this approach is not applicable in our problem setting. Because of the non-zero production lead time, the retailer and manufacturer make their replenishment decisions at different times, using different available information and uncertainty levels. Specifically, while the retailer optimizes her order quantity by considering her inventory and the demand distribution of end customers, the manufacturer must optimize his production decision one period in advance, considering his inventory level and the retailer's order distribution. As a result, simply summing the retailer's and manufacturer's objective functions would not be practical in this setting. Instead, we must seek the optimal decision for a centralized decision-maker that maximizes the profit of the entire supply chain:

Proposition 1. *From a centralized decision-maker perspective, the optimal strategy is that the manufacturer produces the exact quantity ordered by the retailer so that the manufacturer:*

(a) Keeps no inventory and faces no lost sales/holding costs,
(b) Provides the highest MLOR level ($n = N$) to the retailer.

Proposition 1 illustrates that from the centralized decision-maker's perspective, the manufacturer's production and the retailer's ordering decisions should be merged into a single decision, with the aim of optimizing the inventory level at the retailer's depot based on the final customers' demand. Proposition 1(a) further reveals that such a production decision benefits the manufacturer by minimizing costs as it eliminates the need to maintain inventory, resulting in reduced holding and lost sales costs. Proposition 1(b) emphasizes a crucial point. By integrating the manufacturer's production and the retailer's ordering decisions, the manufacturer faces no uncertainty, enabling the precise production of the exact quantity required by the retailer. Consequently, all freshly produced products are shipped directly to the retailer's depot, resulting in the highest possible MLOR level ($n = N$) provided to the retailer. This counterintuitive result challenges the notion that reaching coordination under a strict MLOR agreement might not be feasible. In fact, Proposition 1(b) illustrates that coordinating the supply chain can lead to retailers receiving the highest possible freshness level from manufacturers without imposing any MLOR agreements.

Fig. 5 represents the sequence of events of the centralized decision-maker model in any period t: (1) Receiving the production quantity of the previous period, $p_{c,t-1}$, at the retail depot and updating the inventory, (2) making the production decision $p_{c,t}$ that becomes available at the retail depot at the beginning of the next period $t + 1$, (3) demand realization, and (4) receipt of freshly produced products by the manufacturer and shipping them to the retailer.

Given the one-period production lead time, a centralized decision-maker optimizes the production quantity $p_{c,t}$ to satisfy demand at the retail depot in period $t + 1$. At the beginning of period t, the actual inventory level for the period $t - 1$, y_{t-1}, and the incoming inventory $p_{c,t-1}$ is known. The ending inventory level in t before the delivery of the production quantity, however, is uncertain and can be determined by $Y_t = y_{t-1} + p_{c,t-1} - D_t$. The objective function of the centralized decision-maker can be simplified by:

$$\Pi_{c,t}(p_{c,t}) = r\mathbb{E}\left(\min\left\{Z_t, p_{c,t}\right\}\right) - \beta_r\mathbb{E}\left(Z_t - p_{c,t}\right)^+ - cp_{c,t}. \tag{14}$$

This involves calculating the distribution of an auxiliary random variable Z_t, which can be obtained through the convolution of the two random variables Y_t and D_{t+1}, specifically, $Z_t = Y_t - D_{t+1}$ with corresponding p.d.f. $k_t(\cdot)$ and c.d.f. $K_t(\cdot)$, determined by convolution. In Eq. (14), the first, second, and third terms represent the expected revenue, the expected cost of goodwill at the retailer, and the production cost at the manufacturer facility, respectively. Based on Eq. (14), the optimal time-dependent production decision of the centralized decision-maker model is:

$$p^*_{c,t} = K_t^{-1}\left(\frac{r + \beta_r - c}{r + \beta_r}\right). \tag{15}$$

For effective coordination in the decentralized SC (our baseline scenario), it is crucial that decisions align with those of the centralized decision-maker. As contracts are widely used to facilitate coordination in the SC, in Section 6, we explore various coordinating contract designs suitable for an SC dealing with perishable products with lost sales and positive lead time. However, the primary focus of this section is to study the impact of coordination on waste and profit in the coordinated SC. Hence, we begin by analyzing waste and profit in the coordinated SC.

Waste perspective

This section aims to evaluate the effect of coordination on waste in the SC. To do so, we compare the waste in our baseline scenario (decentralized SC), where the retailer requires the manufacturer to send only freshly produced products ($n = N$), with the waste under the described centralized decision-making model. The waste function of the retailer and the manufacturer under $n = N$ has been introduced earlier in Eqs. (11) and (12). Similar to Section 4, the issuing policy at the retailer is LIFO; therefore the expected waste of the centralized decision-maker model ($\theta_{c,t}$) in each period is:

$$\theta_{c,t} = \mathbb{E}\left(y^1_{c,t} - \left(D_{t+1} - p^*_{c,t} - \sum_{j=2}^{N-1} y^j_{c,t}\right)^+\right)^+. \tag{16}$$

Looking at p^*_t, q^*_t, and $p^*_{c,t}$ in Eqs. (9), (4), and (15) respectively, it is clear that each optimization is based on a different distribution that makes it very difficult to derive any analytical comparison. However, some interesting effects can be shown even numerically.

Fig. 6 contrasts the total waste in the coordinated supply chain with the baseline scenario. This comparison is carried out under various issuing policies and considers both low and high wholesale prices, with the low wholesale price closer to the retail price ($c \ll w < r$) and the high wholesale price closer to the production cost ($c < w \ll r$). Waste in the baseline scenario (decentralized SC) is shown to be consistently higher than in the coordinated SC across all issuing policies. Note that in both cases, the highest possible MLOR level has been provided to the retailer (see Proposition 1). In particular, when the wholesale price is rather high ($c \ll w < r$), the waste increases in the baseline scenario. This is attributed to the retailer's lower marginal profit and reduced willingness to order when wholesale prices are higher. However, manufacturers are inclined to produce more because of their higher marginal profit, leading to increased waste on the manufacturer's side and ultimately throughout the entire supply chain. In contrast, the coordinated model remains insensitive to wholesale price fluctuations since it optimizes the SC from the central decision-maker's viewpoint. Fig. 6 additionally illustrates that although coordination proves effective in reducing waste, regardless of changes in the wholesale price, the magnitude of waste reduction is highly dependent on the issuing policy. Under the LIFO issuing policy, the waste generated by the coordinated SC increases significantly. The main reason for such an effect is that in the coordinated SC, the centralized decision-maker seeks to keep all inventories at retail shops (as close as possible to the final consumers), which results in a more sensitive model regarding the issuing policy. Fig. 6 also shows that, unlike the coordinated SC, the waste produced

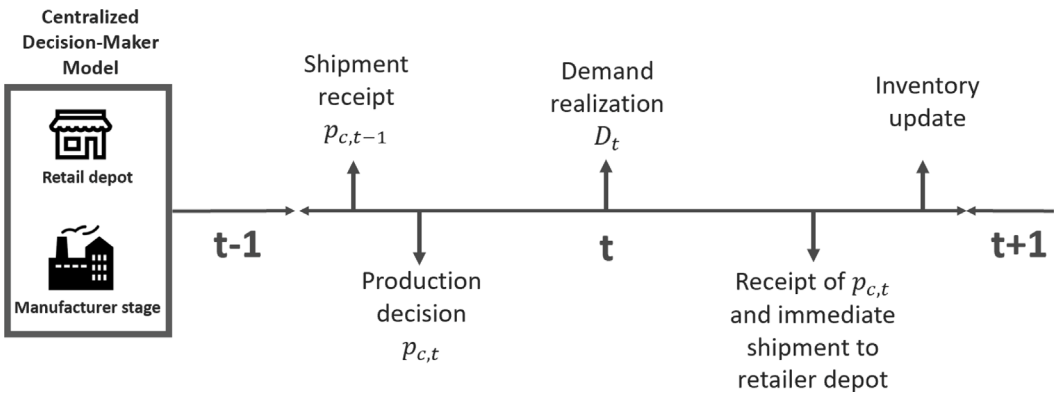


Fig. 5. The sequence of events for the benchmark model in each period.

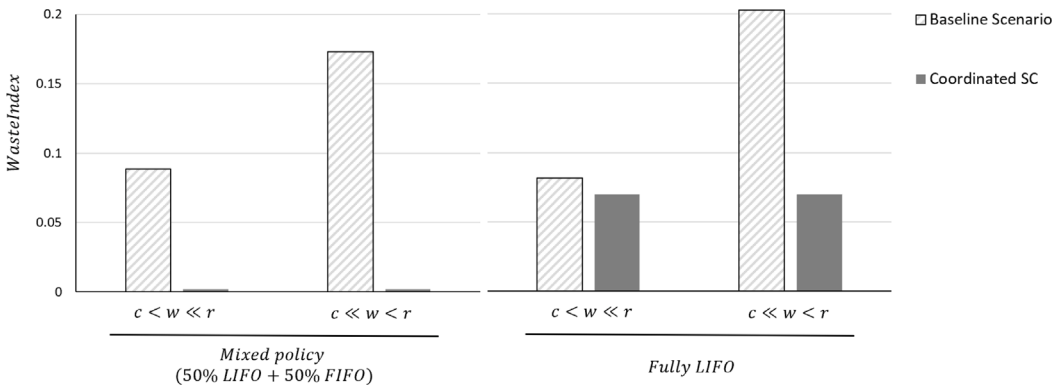


Fig. 6. Changes in the SC's waste by coordination under different issuing policies and wholesale prices.

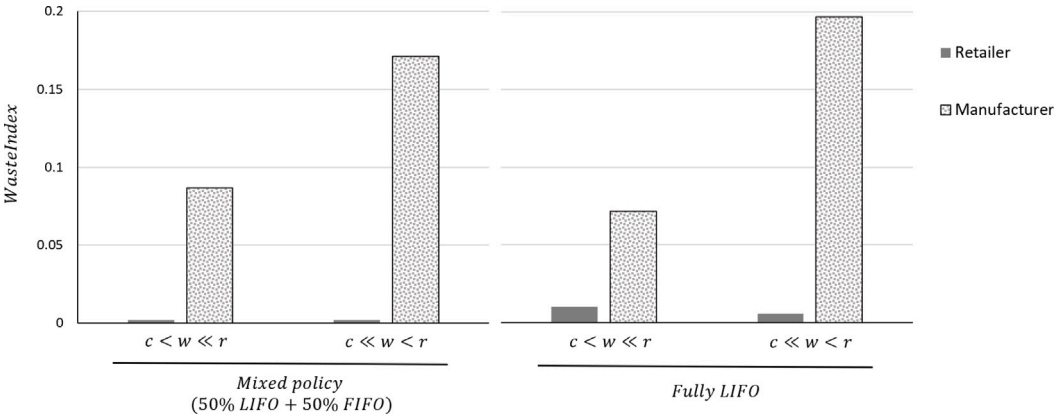


Fig. 7. Waste distribution in the baseline scenario under different issuing policies and wholesale prices.

in the baseline scenario drops under the LIFO issuing policy. This is surprising since one expects the LIFO policy always negatively to affect the waste output.

Fig. 7 illustrates the waste distribution between the retailer and the manufacturer and highlights that regardless of the wholesale price and the issuing policy, in the decentralized SC, the manufacturer is the main contributor to the waste due to his tight MLOR requirement. When different issuing policies are compared, the waste of the retailer increases under the LIFO issuing policy. The increased waste at the retailer's end necessitates higher order quantities to achieve the desired inventory level. These higher-order quantities help the manufacturer sell more and consequently reduce waste. Although the retailer's waste increase is marginal, the manufacturer's reduction is substantial; therefore, overall waste drops under the LIFO issuing policy. It is crucial to emphasize that in our baseline scenario (decentralized SC) and centralized decision-maker model, the manufacturer fulfills orders with freshly produced products ($n = N$). The key distinction lies in eliminating inventory at the manufacturer's end in the centralized model. By optimizing the inventory level at the retailer and eliminating inventory at the manufacturer, the centralized model significantly enhances efficiency and reduces waste in the system.

Profit perspective

Fig. 8 confirms the results of previous studies by showing that switching to a coordinated SC at different wholesale prices leads to greater profit for the SC. Note that the SC's profit under higher wholesale prices is slightly lower due to the manufacturer's higher waste rate,

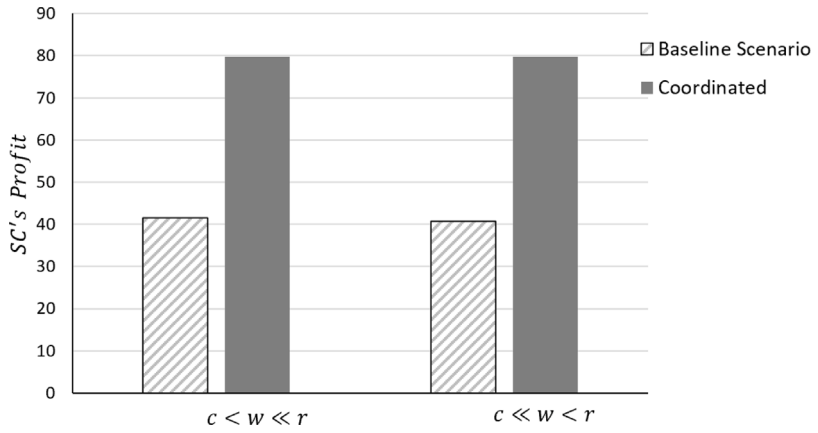


Fig. 8. Profit in the SC under low and high wholesale prices.

which we explained earlier. Therefore, the coordination strategy can also reduce SC waste and increase profit.

6. Coordinating contracts

In this section, we are interested in contracts that provide our baseline scenario (decentralized SC) with the optimal decisions of the centralized decision-maker model in the long run. From the objective function of the centralized decision-maker model (Eq. (14)), we know that the centralized decision-maker faces only one stochastic decision ($p_{c,t}$) (this is known in the literature as a single-location problem). However, in the baseline scenario, the retailer faces demand uncertainty and the manufacturer faces uncertainty about the retailer's order quantity (known as a two-location problem in the literature). The next proposition illustrates that to coordinate the SC, it is required to change the two-location problem to a single-location one.

Proposition 2. *Coordinating the decentralized SC is only possible if the contract design eliminates the uncertainty of the retailer's order for the manufacturer; therefore, Proposition 1 is fulfilled.*

Proposition 2 highlights that by eliminating the retailer's order uncertainty, the manufacturer only produces the required quantity and does not keep any inventory. Based on Proposition 2, to reach coordination, the retailer's order and the manufacturer's production decisions must be fully aligned, and therefore, the manufacturer transfers all freshly produced products to the retailer. Therefore, the highest level of MLOR agreement is met without implementing any MLOR requirements, as discussed in Proposition 1. Two possible approaches to satisfy Proposition 2 and reach coordination, in such a positive lead time and lost sale setting, are to have scanned-based trading or private-label-production agreements between the retailer and manufacturer. In the following, we discuss the possible coordination contracts when the SC is operating under either possible approach.

Scanned-based-trading

Scanned-based-trading (SBT) agreements are becoming popular as a substitute for vendor-managed-inventory (VMI) agreements (Choi, Rabinovich, & Richards, 2019). The main difference is that in the VMI setting, the retailer delegates replenishment decisions to the manufacturer. At the same time, SBT agreements go further, with the manufacturer retaining ownership of products and associated costs until they are sold to consumers (see, e.g., Choi et al., 2019). Thus, in the SBT setting, the manufacturer optimizes the retailer's ordering decisions and takes responsibility for inventory management and related costs. On the one hand, this kind of agreement can be interesting for retailers who work with not well-established products or manufacturers (e.g., new entrepreneurs and start-ups). In these cases, retailers might attempt to minimize their risk of loss and avoid taking ownership of products. On the other hand, SBT is also beneficial for the manufacturer because it prevents the retailer from dealing with the uncertainty of the retailer's order and, as a result, produces as much as needed in the retail store. Therefore, SBT can satisfy the requirement mentioned in Proposition 2 for coordination. However, a proper contract is required to ensure that the manufacturer produces the same amount as the centralized decision-maker model to reach coordination. In the SBT setting, where the manufacturer retains ownership of the products until they are sold to consumers, the retailer does not pay a wholesale price to the manufacturer until the products are sold. As a result, most well-known contract designs (e.g., revenue sharing and buyback) in which the retailer pays a wholesale price for delivered products do not fit into this structure and do not coordinate this problem.

Another form of coordinating contract for the SBT setting involves the retailer receiving a share of the revenue without paying any wholesale price (see, e.g., Choi & Guo, 2020). The consignment contract is a well-known example of such an arrangement (see, e.g., Hu & Feng, 2017; Zhang, Matta, & Lowe, 2010). Renowned retail platforms such as Amazon have utilized this contract, enabling various manufacturers to directly sell their products while paying a portion of their revenue and covering inventory operating costs. However, in these contract types, the decrease in the manufacturer's marginal profit compared to the centralized decision-maker model results in a reduction in the optimal production level, making coordination unachievable. The only remaining possibility of achieving coordination under the SBT agreement is when the retailer only demands a fixed profit from the manufacturer. An example of such a contract is a two-part tariff contract comprising a wholesale price and a fixed side payment. In some industries in China, retailers commonly request such a fixed payment along with a share of the revenue to allow manufacturers to use the provided selling channel (Shen, Willems, & Dai, 2019). Here, we study a two-part tariff contract in which the manufacturer does not charge the wholesale price, but the retailer solely requires a side payment per period from the manufacturer. Since the manufacturer owns the products until they are sold, he is responsible for all operating and penalty costs associated with inventory excess and shortage. Requiring the manufacturer to pay a fixed side payment relieves the retailer of all risks associated with selling the products. This contract setting may be interesting for retailers working with new grocery brands, especially in the organic food industry. Using Eq. (17), the retailer, as the Stackelberg leader, maximizes her profit by setting the side payment (χ) while ensuring that the manufacturer earns his reservation profit (π_m):

$$\hat{\Pi}_r^* = \max \chi \qquad s.t.\ \ \mathbb{E}\hat{\Pi}_m^* = \lim_{T\to\infty} \frac{\sum_{t=1}^{T} \Pi_{m,t}(p_t)}{T} \geq \pi_m. \tag{17}$$

The manufacturer's profit function and the optimal time-dependent production decision are:

$$\Pi_{m,t}(p_t) = r\mathbb{E}\left(\min\left\{Z_t, p_t\right\}\right) - \beta_r \mathbb{E}(Z_t - p_t)^+ - cp_t - \chi, \tag{18}$$

and

$$p_t^* = K_t^{-1}\left(\frac{r+\beta_r-c}{r+\beta_r}\right), \tag{19}$$

respectively. Comparing Eqs. (15) and (19) reveals that using this specific two-part tariff contract, the decentralized SC is coordinating by achieving the same production level as the centralized decision-maker.

Private-label-production

Another possibility of coordinating such a positive lead-time/lost-sales SC is when there is a private-label-production (PLP) agreement between the retailer and the manufacturer, where the retailer is responsible for production decisions. This agreement ensures that the manufacturer and the retailer agree on the production quantity in advance or the retailer announces the decision to the manufacturer. Therefore, the retailer has to cope with the production lead time and announce the required production quantity to the manufacturer in advance. Like the SBT agreement, the requirements mentioned in Proposition 2 are satisfied since the manufacturer is not subject to any uncertainty. Some use case examples include retailers aiming to provide a high service level with reasonable prices, those who do not want any deviations from the order sizes, or those who want to avoid sharing any information regarding their inventory and demand forecast with their manufacturer.

Using PLP agreements, the manufacturer benefits from having no uncertainty, and the retailer ensures that the required quantity of freshly produced products is obtained. By having the production decision made for the manufacturer, the problem changes to a stylized newsvendor problem in which the retailer expects the manufacturer to be fully reliable by fulfilling orders in full. This problem has been extensively studied in both single- and multi-period settings (see, e.g., Bai, Chen, & Xu, 2017; Cachon, 2003; Cai, Chen, Xiao, & Xu, 2010; Chung & Erhun, 2013; Zhang, Liu, Zhang, & Bai, 2015). In addition, similar problems to our PLP setting problem can be found in the advanced ordering literature. In this stream of literature, researchers try to design incentive plans for retailers to convince them to order in advance and reduce uncertainty for manufacturers (see, e.g., Huang, Su, & Lin, 2011). Note that most coordination contracts in the literature require optimizing the parameters for coordination. For example, revenue-sharing contracts need to optimize the manufacturer's revenue portion. However, these contracts may not guarantee the manufacturer's reservation profit. To address this, a side payment can be introduced from the retailer to the manufacturer. Studies have shown that the two-part tariff contract can outperform other coordinating contracts (Lee & Rhee, 2010; Li & Liu, 2015, 2020).

7. Discussion: Coordinate or collaborate?

We have shown that both coordination and collaboration strategies can effectively reduce waste and improve profit in perishable-product SCs. In this section, we compare the performance of the two strategies and advise practitioners on the best course of action. Fig. 9 compares the collaboration and coordination strategies in terms of total profit and waste in the SC under different issuing policies. It is clear from Fig. 9(a) that the coordination strategy generates more profit for the SC compared to the collaboration one. In the coordination strategy, the manufacturer is convinced to produce more in each period due to a higher marginal profit. This leads to more inventory on the retail side, resulting in higher sales than in the collaborative strategy. A similar dynamic unfolds for the collaboration strategy: Due to a higher wholesale price, the manufacturer has a higher profit margin and produces more products. However, having a higher inventory on the retail side results in more aged products on the shelves over time, especially when customers are very picky about choosing the freshest products (LIFO).

Fig. 9(b) shows that the coordination strategy generates significantly more waste than collaboration due to the issuing policy leaning toward LIFO. Although the advantage of coordinating the SC is a higher profit for the entire SC, the waste produced in this strategy is highly sensitive to the issuing policy compared to the collaboration strategy. Therefore, when retailers have less power to influence the issuing policy by selling the least fresh products first (such as brick-and-mortar grocery stores), the collaboration strategy has a clear advantage in terms of reducing waste throughout the SC. However, for online retailers who choose which products to sell first, coordination might be a better choice to reduce waste in the SC. Fig. 9 illustrates that the best overall option for a two-echelon SC chain of perishable products is to coordinate the SC to have a higher profit while ensuring that the mixture of the issuing policy is more FIFO than LIFO to minimize waste as much as possible.

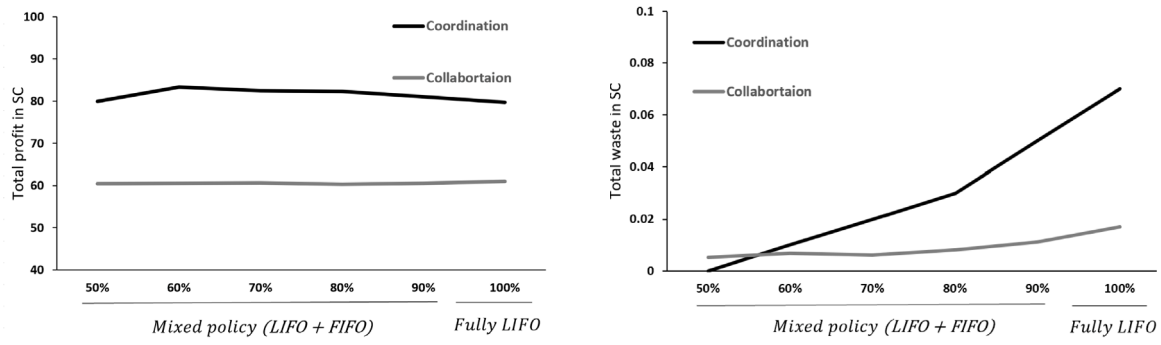
8. Conclusion

Reducing food waste supply chains with multiple independent operating firms is challenging. For perishable products, retailers typically demand a high remaining shelf life upon receipt from manufacturers (known as the MLOR agreement). While a higher MLOR level increases the handling window and higher service levels for retailers, manufacturers encounter the opposite. For manufacturers, a high MLOR level, due to short internal shelf life, creates many hurdles to fulfilling retailers' uncertain orders in full, especially when they have a positive production lead time. Retailers often assess manufacturers based on "on time in full" performance, leading manufacturers to prefer overstock to stockouts, potentially contributing to significant food waste. To address this inefficiency, we study two strategies for retailers and manufacturers to follow: collaboration on setting the MLOR level and coordination of the SC. We investigate their impact on the SC's profit and waste compared to our baseline scenario, where the retailer requires the manufacturer to send only freshly produced products.

First, we investigate the effect of collaborating on the MLOR level, which means that the retailer also allows the manufacturer to send products with a lower shelf life. In terms of waste, we analytically show that reducing the MLOR by one period decreases the manufacturer's average waste in the long run. For the retailer, although the long-run average waste increases, we demonstrate that it is bounded. We further show that when the shelf life is large enough, waste in the SC decreases by requiring the manufacturer to adopt a lower MLOR level. Specifically, we show that under lower MLOR levels, the long-run average waste of the manufacturer decreases to almost zero, while the increase in retailer waste is negligible.

With regard to profits, we analytically demonstrate that the manufacturer is capable of accepting a lower wholesale price if it is not only obliged to send freshly produced products. Specifically, the retailer's profit increases as a result of a lower wholesale price at a lower MLOR level. Our numerical result supports this analytical result, even under the relaxed assumption made for analytical tractability. It affirms that the higher the manufacturer's reservation profit, the more profitable it is for the retailer to reduce the MLOR to achieve a lower wholesale price. In particular, we also numerically demonstrate that reductions in the wholesale price and the MLOR level positively influence the efficiency of SC, even if this increase is insufficient to fully coordinate SC. In studying the impact of coordination on the SC's waste and profit, we found that the waste of the baseline scenario is higher under higher wholesale prices. At the same time, the coordinated SC is insensitive to the wholesale price. Moreover, we illustrate that coordination is beneficial in reducing waste in the SC, while the magnitude of the effect depends on the issuing policy. This higher sensitivity regarding the issuing policy is due to the policy compelling the retailer to keep higher inventory, compared to the baseline scenario, to increase sales.

Furthermore, our baseline scenario (decentralized SC) shows that the manufacturer is the main contributor to food waste in the supply chain. With regard to profit, our results confirm that coordination



(a) Total profit in the SC

(b) Total waste in the SC

Fig. 9. Comparing collaboration and coordination strategies.

increases profit regardless of the wholesale price. Thus, the coordination strategy can reduce waste and increase profits in the SC. Our findings furthermore reveal that coordinating the SC ensures that the manufacturer supplies only freshly produced products to the retailer, resulting in the highest possible freshness level for the retailer's inventory. This is particularly intriguing as it allows retailers to benefit from the high freshness provided by manufacturers without imposing any restrictive MLOR agreement on them. We further discuss possible coordination contracts for such a positive lead time lost sales decentralized SC. Our study highlights that coordination is possible when the retailer and manufacturer have either scanned-based-trading or private-label-production agreements.

From a managerial point of view, focusing solely on reducing waste in retail stores (like when governments set a fine on the quantity of retail waste) might not be the best policy for reducing waste in the SC. These policies only motivate retailers to raise the MLOR level to protect their own profits, reduce waste, and guarantee their customers' satisfaction, while it has a counter effect on manufacturers. Setting the MLOR level too high strongly affects the manufacturers' waste and reduces their ability to fulfill orders in full. Our results show that the best MLOR level is when manufacturers have some flexibility in fulfilling the retailers' orders considering transportation lead time. Lowering the MLOR level helps manufacturers save some costs while offering a lower wholesale price, which can be used as an incentive to encourage retailers to collaborate in setting the MLOR level. Ultimately, collaboration leads to an increase in profit and a significant reduction in the waste of grocery SCs.

Furthermore, while coordination can reduce inefficiency and increase profit in the SC, it might result in higher waste volume than the collaboration strategy. To reduce waste in brick-and-mortar grocery retailers, where the issuing policy is more LIFO than FIFO, it might be better to collaborate on setting the MLOR level rather than coordinating the SC. However, for online groceries, which have strong control over the issuing policy, coordination might be an excellent choice, as it reduces waste and increases profit in the SC while ensuring the highest freshness possible for retailers. Although the industry's general tendency has been for retailers to assume the burden of reducing food waste, our study contributes to a nascent understanding that food waste at the manufacturers' level needs to be a part of retailers' environmental considerations. Therefore, novel contracting approaches can contribute to more sustainable food SCs.

Our study faces a couple of limitations. First, we assume that the retailer and the manufacturer optimize their ordering and production quantities following a dynamic and myopic inventory control/production policy, that is, only considering the period that the decision primarily affects. Although this is a limitation, considering the value-to-go functions for both firms would require a stochastic-dynamic programming approach for each stage in the supply chain, i.e., for the retailer and the manufacturer, which is intractable in the coordination setting. A possible policy for future research is the basestock policy, which has been shown to be asymptotically optimal in perishable-product inventory control. Moreover, we only consider a limited number of contracts. An extension could be to study the impact of MLOR agreements considering other types of contracts, such as revenue-sharing or quantity-flexibility contracts. In conclusion, while there is a vast literature in the field of supply chain coordination for products with a limited shelf life, this topic is still understudied and requires more research in the future.

CRediT authorship contribution statement

Navid Mohamadi: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Sandra Transchel:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Jan C. Fransoo:** Validation, Supervision, Investigation, Conceptualization.

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Appendix A

Derivation of Q_{t+1} and its density and distribution function

Note, at the end of period $t+1$, the retailer's order quantity would be $q^*_{t+1} = \left(F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) - \sum_{j=1}^{N-1} y^j_{r,t+1}\right)^+$; however, $\sum_{j=1}^{N-1} y^j_{r,t+1}$, denoting the inventory level at he of period $t+1$ that has to be carried over to $t+2$, is unknown at the beginning of period $t+1$, or equivalently, at the end of period t, when the retailer determines Q_{t+1}. For a simplified notation, let $\tilde{Y}_{r,t+1}$ denote the random inventory level at the end of period $t+1$, feasible to be carried over to period $t+2$. Thus, the random order quantity at the end of period $t+1$ is

$$Q_{t+1} = \left(F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) - \tilde{Y}_{r,t+1}\right)^+. \tag{20}$$

The retailer can determine the distribution of $\tilde{Y}_{r,t+1}$ based on her ending inventory at the end t, specifically, $\tilde{Y}_{r,t+1} = \left(\sum_{j=1}^{N-1} y^j_{r,t} + q_t - D_{t+1} - Y^0_{r,t+1}\right)^+$, i.e., the inventory at the beginning of period $t+1$, including the order quantity from the previous period, q_t, minus the demand, D_{t+1}, and minus the waste at the end of period $t+1$, $Y^0_{r,t+1}$, still unknown

at that time. Since $\sum_{j=1}^{N-1} y_{r,t}^{j} + q_t = F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right)$, it follows:

$$\tilde{Y}_{r,t+1} = \left(F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) - D_{t+1} - Y_{r,t+1}^{0}\right)^{+}. \quad (21)$$

Depending on the retailer's inventory issuing policy (FIFO or LIFO), $Y_{r,t+1}^{0}$ is calculated by

$$Y_{r,t+1}^{0} = \begin{cases} \left(y_{rt}^{1} - D_{t+1}\right)^{+} & FIFO \\ \left(y_{rt}^{1} - \left(D_{t+1} - \sum_{j=2}^{N-1} y_{rt}^{j}\right)^{+}\right)^{+} & LIFO \end{cases} \quad (22)$$

such that

$$Q_{t+1} = \begin{cases} F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) - \left(F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) - D_{t+1} - \left(y_{rt}^{1} - D_{t+1}\right)^{+}\right)^{+} & FIFO \\ F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) - \left(F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) - D_{t+1} - \left(y_{rt}^{1} - \left(D_{t+1} - \sum_{j=2}^{N-1} y_{rt}^{j}\right)^{+}\right)^{+}\right)^{+} & LIFO \end{cases}. \quad (23)$$

One can see that the distribution of Q_{t+1} directly depends on the distribution of D_{t+1}. Furthermore, under FIFO it holds that for all demand realizations $d_{t+1} \le y_{rt}^{1}$, it follows $q_{t+1} = y_{rt}^{1}$, if $y_{rt}^{1} < d_{t+1} < F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right)$ it follows $q_{t+1} = d_{t+1}$, and for $F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) \le d_{t+1}$, it follows $q_{t+1} = F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right)$. With a similar derivation for LIFO, from Eq. (23) we can derive G_{t+1} under FIFO

$$G_{t+1}^{FIFO}(q) = \begin{cases} F(q) & : y_{rt}^{1} \le q < F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) \\ 1 & : q = F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) \end{cases} \quad (24)$$

and under LIFO

$$G_{t+1}^{LIFO}(q) = \begin{cases} F(q - y_{rt}^{1}) & : y_{rt}^{1} \le q < y_{rt}^{1} + \sum_{j=2}^{N-1} y_{rt}^{j} \\ F(q) & : y_{rt}^{1} + \sum_{j=2}^{N-1} y_{rt}^{j} \le q < F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right) \\ 1 & : q = F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right). \end{cases} \quad (25)$$

We find that the distribution of the retailer's order quantity depends on the age distribution of the inventories.

Proof of Theorem 1. We prove this theorem for the case that $n = N$, that is, the manufacturer can only send the freshest products. The proof also holds for $n < N$, but contains more mathematical terms. The optimal production quantity of the manufacturer is $p_t^*(w,N) = \max\left\{G_t^{-1}(\alpha), G_t^{-1}\left(\frac{w-c}{w}\right)\right\}$. Assuming $\alpha \ge \frac{w-c}{w}$, we obtain $p_t^*(w,N) = G_t^{-1}(\alpha)$. From (24) and (25), we can see that if $\alpha \ge \frac{r+\beta_r-w}{r+\beta_r}$ it holds $G_t^{-1}(\alpha) = F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right)$. That is, the manufacturer produces the maximum order quantity of the retailer, that is, $p_t^*(w,N) = F^{-1}\left(\frac{r+\beta_r-w}{r+\beta_r}\right)$ and is therefore always able to fulfill the entire order quantity of the retailer.

Proof of Theorem 2. (a) We can rewrite the manufacturer's optimal production decision function (9) for two specific cases of $n = N-1$ (shown as $p_t(w,N-1)$) and $n = N$ (shown as $p_t(w,N)$) as follows:

$$p_t^*(w,N) = \max\left\{G_t^{-1}(\alpha), G_t^{-1}\left(\frac{w-c}{w}\right)\right\} \quad (26)$$

$$p_t^*(w,N-1) = \max\left\{\left(G_t^{-1}(\alpha) - y_{m,t-1}^{N}\right)^{+}, \times \left(G_t^{-1}\left(\frac{w-c}{w}\right) - y_{m,t-1}^{N}\right)^{+}\right\} \quad (27)$$

Given that the in the long run $y_{m,t-1}^{N}$ is not exceeding the maximum service level provided $\max\left\{\alpha, \left(\frac{w-c}{w}\right)\right\}$, we can conclude from (26) and (27):

$$p_t^*(w,N) = p_t^*(w,N-1) + y_{m,t-1}^{N} = \max\left\{G_t^{-1}(\alpha), G_t^{-1}\left(\frac{w-c}{w}\right)\right\}. \quad (28)$$

Considering (28), one can conclude that $p_t^*(w,N) \ge y_{m,t-1}^{N}$. Knowing this, from (10) and (11), it can be concluded that $\theta_{m,t}(w, n = N-1) \le \theta_{m,t}(w, n = N)$. If in any specific period (e.g., (t) $\theta_{m,t}(w, n = N-1) = \theta_{m,t}(w, n = N)$, this means $p_t^*(N) = y_{m,t-1}^{N}$, and therefore $p_t^*(N-1) = 0$. In this case, the remaining inventory for the next period under $n = N-1$ will be zero ($y_{m,t}^{N} = 0$) therefore $\theta_{m,t+1}(w, N-1) = 0$. In other words, in the succeeding period $\theta_{m,t+1}(w, N-1) < \theta_{m,t+1}(w, N)$. As a result, in the long run, $\theta_m(w, n = N-1) < \theta_m(w, n = N)$.

(b) Based on (10) and (11), the expected waste reduction per period is:

$$\sum_{t=1}^{T}(\mathbb{E}(p_t^* - Q_t)^{+} - \mathbb{E}(y_{m,t-1}^{N} - Q_t)^{+}) = \frac{1}{2}((p_t^*)^2 - (y_{m,t-1}^{N})^2)$$

Therefore, the long-run average waste reduction per period is:

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \frac{1}{2}((p_t^*)^2 - (y_{m,t-1}^{N})^2) = \frac{1}{2}((p_t^*)^2 - (y_{m,t-1}^{N})^2).$$

Proof of Theorem 3. (a) Based on MLOR agreements $n = N$ and $n = N-1$, the ages of arriving shipments are $q_t^*(w,N) = y_{r,t}^{N}$ and $q_t^*(w,N-1) = y_{r,t}^{N} + y_{r,t}^{N-1}$, respectively. To calculate the maximum increase in the average expected waste of the retailer under $n = N-1$, we assume that all arriving shipments have the lowest possible age allowed by MLOR, $q_t^*(w,N-1) = y_{r,t}^{N-1}$. To find the upper bound of the increase in the waste of the retailer, we assume that the realization of demand in each period is zero, and the issuing policy is LIFO. Having the realization of demand equal to zero means that in both cases, all ordered products will ultimately be wasted, whereas the difference is in the number of required replenishments. When $n = N$, every N periods, the retailer orders from the manufacturer an amount of $q_t^* = q_t^*(w,N)$, while when $n = N-1$, every $N-1$ periods, the same amount is ordered $q_t^* = q_t^*(w,N-1)$. Therefore, the upper bound of the increase in the waste of the retailer per period is:

$$\lim_{T\to\infty} \frac{1}{T}\left(q_t^* \frac{T}{N-1} - q_t^* \frac{T}{N}\right) = \lim_{T\to\infty} \frac{q_t^*}{T} \frac{T}{N(N-1)} = \frac{q_t^*}{N(N-1)}.$$

Proof of Theorem 4. (a) From (28), one can see that $p_t^*(w,N-1) \le p_t^*(w,N)$.

(b) We can rewrite (7) as:

$$\Pi_{m,t}(p_t(N-1)) = w\mathbb{E}\left(\min\left\{Q_t, p_t(N-1) + y_{m,t-1}^{N}\right\}\right) - cp_t(N-1) \quad (29)$$

$$\Pi_{m,t}(p_t(N)) = w\mathbb{E}\left(\min\left\{Q_t, p_t(N)\right\}\right) - cp_t(N) \quad (30)$$

Taking into account (28), while the first two terms of the profit functions in (29) and (30) are the same, production costs are $cp_t^*(w,N) \ge cp_t^*(w,N-1)$. Therefore, for any given period, $\Pi_{m,t}(p_t(w,N-1)) \ge \Pi_{m,t}(p_t(w,N))$. Moreover, for any given period (e.g., t-1), under $n = N-1$, when the observed order of the retailer (q_t) at the end of the period is lower than the available inventory of the manufacturer, the manufacturer ultimately has some inventories left:

$$p_{t-1}^*(w,N-1) + y_{m,t-2}^{N} > q_t \quad (31)$$

Based on (31), since the manufacturer uses the FIFO issuing policy, after fulfilling the retailer's order, the updated inventory level at the

end of period $t-1$ will be positive ($y^N_{m,t-1} > 0$). Based on (28), when $y^N_{m,t-1} > 0$, then $p^*_t(w,N) > p^*_t(w,N-1)$. Therefore, $cp^*_t(w,N) > cp^*_t(w,N-1)$, and we can conclude $\Pi_{m,t}(p_t(w,N-1)) > \Pi_{m,t}(p_t(w,N))$.

The previous result holds if the observed order of the retailer (q_t) at the end of the period $t-1$ is lower than the available inventory level of the manufacturer. Since $p^*_{t-1}(w,N-1) + y^N_{m,t-2} = G^{-1}_t(\frac{w-c}{w})$, with a probability of $\frac{w-c}{w}$, the realization of the retailer order is lower than the available inventory level. Therefore, we claim that if in only one period the above condition holds, the profit of the manufacturer over the contract horizon will be:

$$\begin{aligned}\hat{\Pi}_m\left(w,N-1,\mathbf{p}^*(w,N-1)\right) &= \sum_{t=1}^{T}\Pi_{m,t}(p_t(w,N-1)) > \\ &\hat{\Pi}_m((w,N),\mathbf{p}^*(w,N)) \\ &= \sum_{t=1}^{T}\Pi_{m,t}(p_t(w,N)).\end{aligned} \tag{32}$$

As a result, the corresponding probability of (32) is equal to the probability that q_t is lower than the available inventory of the manufacturer at least in one period, which is:

$$\lim_{T\to\infty} 1-\left(1-\frac{w-c}{w}\right)^T = \lim_{T\to\infty} 1-\left(\frac{c}{w}\right)^T = 1.$$

(c) Given an MLOR level $n = N$ for which the retailer, acting as a Stackelberg leader, optimizes the wholesale price $w^*(N)$ so that the manufacturer exactly earns his reservation profit (π_m). Therefore, the $w^*(n)$, of the first-stage problem in Eq. (1), given $\Omega = (w,n)$ is as follow:

$$w^*(n) = \underset{w}{\operatorname{argmin}}(\hat{\Pi}_m((w,n),\mathbf{p}^*(w,n)) - \pi_m) \tag{33}$$

From Eqs. (33) and (32), it is intuitive that $w^*(N) > w^*(N-1)$.

Proof of Proposition 1. (a) Suppose that the central decision-maker keeps inventory at the manufacturer's facility. This means that the production decision should be higher than the retailer's order. Therefore, in case the manufacturer keeps inventories, the production cost and the holding cost of the manufacturer increase while the revenue stays unchanged. Due to this inefficiency, keeping inventory in the manufacturer's facility is not an optimal strategy for the centralized decision-maker.

(b) When the manufacturer does not maintain any inventory and knows the exact order size of the retailer, she produces the exact order size and fulfills the orders with those freshly produced products. In other words, the highest MLOR level is provided to the retailer without imposing any restrictive MLOR agreements.

Proof of Proposition 2. Suppose that the contract is designed so that the problem remains two-location, which means that the retailer faces demand uncertainty and the manufacturer faces the retailer's order uncertainty. As a result, since the manufacturer is faced with a stochastic decision, he might produce less or more than the required quantity. Having fewer products produced means not fulfilling the order in full, while having more associates with extra costs for the manufacturer's facility. In both cases, the supply chain works less efficiently than the centralized one. Therefore, to achieve the same performance as the centralized decision-maker, it is necessary to change the two-location problem to a single-location one. Consequently, the coordination contract should be designed to eliminate the retailer's order uncertainty. As a result, the manufacturer works more efficiently (like the centralized decision-maker), which might lead to coordination.

Table of symbols and used variables

Symbol	Description
N	Shelf life of the product
r	Retail price
w	Wholesale price
c	Production cost
n	MLOR agreement
N	Maximum shelf life
T	Contract horizon
q_t	Order decision of retailer
p_t	Production decision of manufacturer
D_t	Stochastic demand variable of retailer
μ	Mean of D_t
σ	Standard deviation of D_t
$p_{c,t}$	Production decision of the centralized decision maker at period t
Ω	Contractual agreement between retailer and manufacturer
Π_m	Long-run average expected profit per period of manufacturer
Π_r	Long-run average expected profit per period of retailer
π_m	Reservation profit of the manufacturer
β_r	Cost of goodwill of retailer
$y^j_{r,t}$	Retailer's inventory with a remaining shelf life of j periods at the end of period t
$y^j_{m,t}$	Manufacturer's inventory with a remaining shelf life of j periods at the end of period t
α	Required service level of the retailer from the manufacturer
Q_t	Retailer's order distribution
$Y^0_{r,t+1}$	Retailer's inventory at the end of period $t+1$ with 0 remaining shelf life
$\theta_{r,t}$	Expected waste at retailer at end of period t
$\theta_{m,t}$	Expected waste at manufacturer at end of period t
$\theta_{c,t}$	Expected waste at Central decision maker at end of period t
$WasteIndex_m$	Long-run average expected waste of the manufacturer
$WasteIndex_r$	Long-run average expected waste of the retailer
Z_t	Demand distribution used by central decision maker give D_t
η^j_t	Shipment sent by the manufacturer with the remaining shelf life of j at the end of period t

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